nature food

Article

https://doi.org/10.1038/s43016-024-01093-9

China's municipal wastewater policies enhanced seafood safety and offset health risks from atmospheric mercury emissions in the past four decades

Received: 14 March 2024

Accepted: 13 November 2024

Published online: 2 January 2025

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The neurotoxin methylmercury in seafood threatens food safety worldwide. China has implemented stringent wastewater policies, established numerous treatment facilities and enforced rigorous water quality standards to address pollution in its waterways. However, the impact of these policies on seafood safety and methylmercury exposure remains unknown. Here we developed a process-based model showing that, although mercury reductions from municipal wastewater policies accounted for only 9% of atmospheric mercury emissions during 1980–2022, these measures unexpectedly prevented $102,000^{+11,000}_{-6,600}$ mercury-related deaths and counteracted nearly two thirds of potential deaths from those emissions. Furthermore, these policies ensured that 146^{+8}_{-9} megatonnes of freshwater seafood met the World Health Organization and China's mercury-safety standards, preventing US498^{+32}_{-29} billion in economic losses. Finally, we explore how China, as the primary global seafood producer and exporter, could develop municipal wastewater policies at the regional level to reduce aquatic pollutants and unlock the health benefits of seafood consumption.

Seafood is integral to global nutrition, economy and cultural practices, with a growing reliance on freshwater seafood production[1,2]. Rich in vital micronutrients, such as omega-3 fatty acids, vitamins and minerals, seafood contributes to the dietary needs of 3.1 billion individuals, supplying approximately 20% of their dietary animal protein[3,4]. Nevertheless, concerns about mercury (Hg) intake have cast a shadow over seafood consumption[5,6]. Hg in aquatic ecosystems can transform into methylmercury (MeHg), a potent neurotoxic substance intricately linked to developmental delays and long-term neurocognitive deficits in fetuses and cardiovascular diseases in adults[7–9]. Because of its high bioavailability and biomagnification, MeHg substantially accumulates in aquatic food webs, leading to alarming levels in seafood[10,11]. Since seafood consumption contributes to over 80% of human MeHg exposure worldwide, reducing its level is essential to bolster consumer confidence and safeguard health[12,13].

China is the largest source of anthropogenic Hg releases worldwide, with notable contributions to both air and water[14]. In recent years,

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https://doi.org/10.1038/s43016-024-01093-9

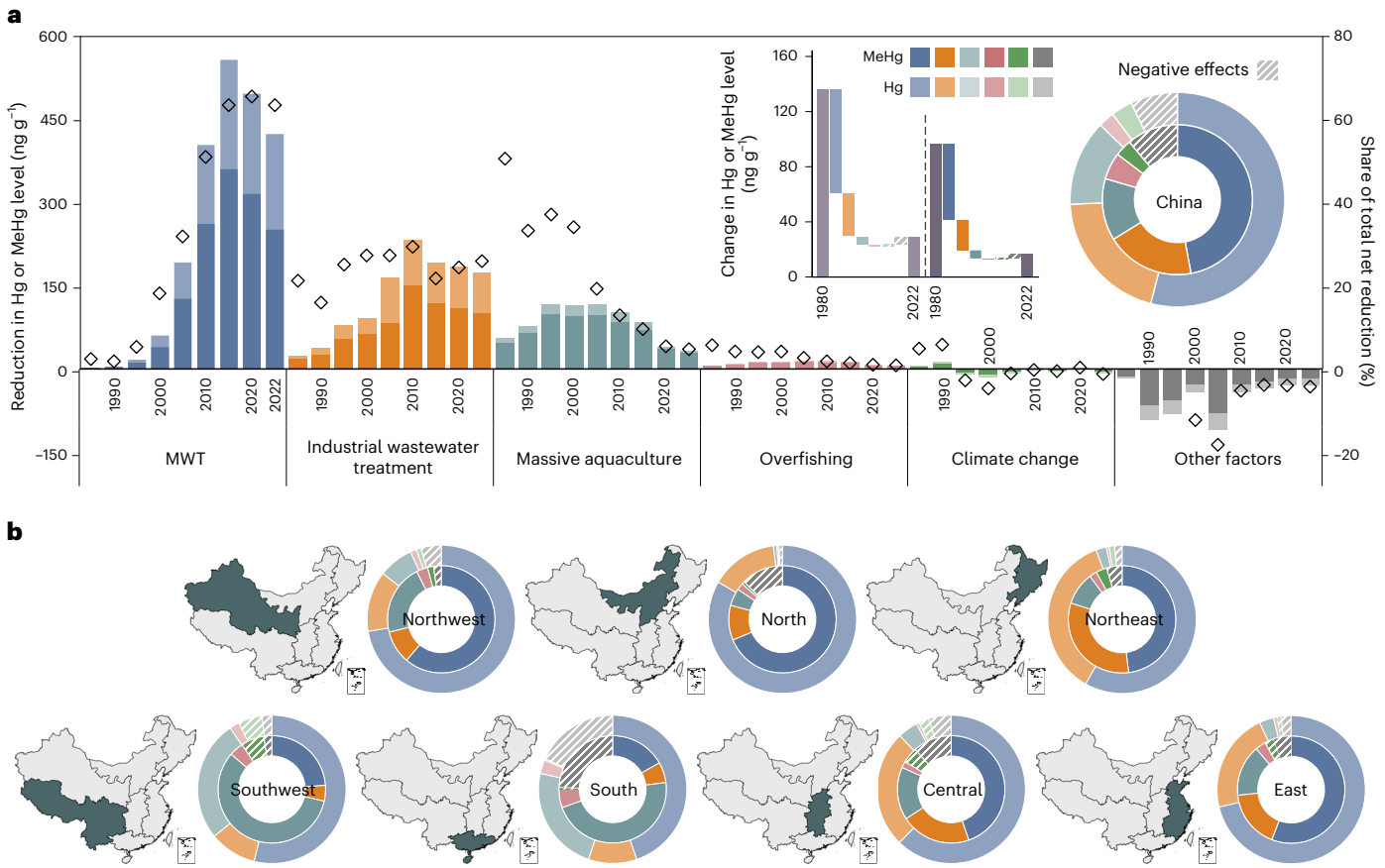


Fig. 1 | Reductions in Hg levels in China's freshwater seafood contributed through major influential factors during 1980–2022. a, Net reductions in Hg levels. The diamonds indicate annual shares of the factors in total net reduction (right *y* axis). Right inset: the pie chart shows the absolute average shares of the factors in total net reduction. Negative impacts are denoted with hatching. Other factors involve atmospheric deposition, surface sinkage, soil erosion, fugitive releases and other natural processes. Left inset: the waterfall plots show the total contributions of influential factors during the decades. **b**, Regional absolute average shares of the factors in total net reduction. The pie charts show the absolute average shares of the factors in total net reduction. Base map in **b** from https://www.tianditu.gov.cn/.

municipal wastewater has been a key source of aquatic Hg pollution, responsible for a substantial portion of Hg released in China[15] and globally[16]. Meanwhile, China is the predominant provider of aquaculture and capture fisheries (60% and 15%, respectively), with seafood exports accounting for 14% of the global total[17]. Given the worldwide elevated fish consumption rates[18], China plays an important role in global seafood safety, which is crucial for food security and public health. Since the 1980s, China has greatly enhanced the adoption and implementation of its wastewater treatment policies, particularly in municipal wastewater treatment (MWT), to improve urban sanitation and alleviate water pollution, especially eutrophication. These initiatives have included constructing centralized treatment facilities, implementing stringent water quality standards, and improving the effluent's quality and the efficiency of MWT. As a result, numerous MWT facilities have been established, with wastewater collection rates jumping from nearly zero to an impressive 98%[19,20] (Supplementary Data 1).

Although initially not focused on Hg[21], recent studies have noted that these MWT policies have contributed to a substantial national reduction in Hg releases into aquatic environments[15,22]. Given that aquatic ecosystems are the principal conduit through which fish and shellfish assimilate and accumulate Hg, the development of MWT policies is expected to reduce Hg levels in China's fish populations, expand Hg-safer seafood production and fortify human health protections by preventing excessive MeHg intake from seafood consumption. However, most current models assessing trace toxicants focus only on the impact of atmospheric emissions on public health[23–25], neglecting pathways such as point-source release into aquatic environments[26]. The effectiveness of China's four-decade-long wastewater policies in enhancing seafood safety and reducing Hg exposure thus remains unclear. Addressing this issue is important, as combating Hg pollution and ensuring food safety are of global concern, and these policies could serve as valuable benchmarks for many developing countries.

To address this knowledge gap, we developed an entire chain of biogeochemical simulations from aquatic Hg release to human risks by outlining an array of model components, including Hg release inventories, dynamic material flow analysis, river network models, bioconcentration simulations, ecological risk assessments, interregional transport models and health impact assessments. Specifically, we developed unit-based datasets for MWT plants, physical transport matrices for freshwater seafood and species-specific bioconcentration simulation approaches in China during 1980–2022, addressing technical challenges faced in previous studies. By incorporating observations of Hg levels in freshwater seafood, we first analysed the impact of China's wastewater policies on the trend in Hg levels in various freshwater seafood species over the past four decades. We then examined the effects of major theoretical factors such as aquaculture expansion, overfishing, climate change, atmospheric deposition, surface sinkage, soil erosion and fugitive release on freshwater seafood Hg levels over the past four decades, comparing these effects with those of wastewater treatment. We also evaluated the substantial environmental, economic and health impacts of this trend in seafood Hg levels. Finally, we explored the broader implications of China's wastewater policies for seafood safety and human health.

Results and discussion

Substantial reduction in seafood Hg due to MWT policies

A key finding of our study is that municipal and industrial wastewater treatment policies have successfully decoupled China's atmospheric Hg emissions and seafood Hg contamination. China's atmospheric Hg emissions have almost quadrupled over the past four decades[27]. However, Hg levels in the country's seafood have declined by 70% during this period[28]. Our analysis indicates that the development of MWT alone explains 71% of the reduction in Hg levels in freshwater seafood (Supplementary Text 1). As illustrated in Fig. 1a, the impact of the MWT policies on Hg levels became particularly notable after 2000, following China's ambitious initiative to establish secondary MWT facilities across all cities before 2010 (ref. 29).

By 2022, the MWT industry had reduced Hg and MeHg levels in freshwater seafood by 421 and 250 ng g^{-1} wet weight, respectively. However, without the implementation of MWT policies, the Hg and MeHg levels in China's freshwater seafood would have reached 592 ng g^{-1} and 289 ng g^{-1}, respectively, in 2022, equal to seven to eight times the current levels (Supplementary Data 2). The most substantial reductions were observed in North and East China, where Hg levels in freshwater seafood decreased by 611 and 531 ng g^{-1}, respectively (Fig. 1b). In contrast, the impacts of MWT in South and Southwest China were minimal, at 61 and 123 ng g^{-1}, respectively. Beyond infrastructure development, these regional disparities are largely due to differences in MWT efficiency. Regions with dense urban populations or arid climates tend to achieve higher Hg concentrations in municipal wastewater, leading to elevated removal rates through MWT[30].

Although less effective than MWT, industrial wastewater treatment had reduced Hg and MeHg levels in freshwater seafood by 172 and 100 ng g^{-1} by 2022, respectively—approximately 40% of the impact achieved by MWT (Fig. 1a). Among the regions, Northeast China, a hub of heavy industry, has witnessed notable benefits from industrial wastewater policies in reducing Hg contamination in freshwater seafood (Fig. 1b). The national focus on industrial wastewater treatment, particularly during China's Ninth Five-Year Plan (1995–1999), was crucial. Policies such as pollution cap control and supervisory monitoring established on-site pretreatment to control toxic discharges, including heavy metals. The introduction of the Pollutant Discharge Permit System in 2004 and the Law on Prevention and Control of Water Pollution in 2008 further strengthened these efforts. Nevertheless, a slight decline in the contribution of industrial wastewater treatment has been observed since 2010 (Fig. 1a), probably due to the avoidance of Hg-associated industrial processes and reduced consumption of Hg-containing products following the implementation of the Minamata Convention on Mercury[31–33]. Another decline during 2020–2022 was plausibly attributed to increased residential water usage and decreased industrial activity during the COVID-19 pandemic[34].

Recent research has hypothesized that the growth of the aquaculture industry may be the primary driver behind the substantial reduction in Hg levels in freshwater seafood in China[35]. However, our findings indicate that the contribution of aquaculture to the reduction of Hg and MeHg in freshwater seafood is relatively small (31 and 29 ng g^{-1} in 2022, respectively), equivalent to the 7.3% and 11.6% of the reductions achieved by the MWT. This unexpectedly small contribution was presumably due to the insensitivity of the bioconcentration ability to aquaculture growth. The most pronounced decreases in Hg and MeHg due to aquaculture were observed in South China and Southwest China, probably because of the expansion of aquaculture following the 1978 Reform and Opening Up Policy, which prioritized the exploitation of inland water resources for aquaculture (Fig. 1b)[36]. Additionally, overfishing has led to smaller reductions in Hg and MeHg levels in freshwater seafood (6.0 and 4.8 ng g^{-1} in 2022, respectively), although it shortens the lifespan of wild fish, lowers their trophic level and reduces accumulated Hg. Freshwater aquaculture now accounts for approximately 85% of the freshwater seafood production in China[17]; the effects of overfishing on Hg levels were therefore essentially 'enclosed.'

Our research suggests that the effects of climate change, represented in our model by changing rainfall and surface runoff, neutrally impacted Hg levels in freshwater seafood (Fig. 1a). This conclusion is not definitive, and further studies on the influence of climate change are necessary (Supplementary Text 2). Other contributors, such as atmospheric deposition, surface sinkage, soil erosion and fugitive releases, have collectively elevated Hg levels in freshwater seafood over the past four decades. This period saw a rapid expansion in Hg-emitting industries, intensifying Hg contamination of seafood in China[37]. Nevertheless, these adverse impacts were overshadowed by the positive outcomes of wastewater treatment initiatives, implying that the implementation of wastewater treatment policies is crucial for mitigating the severe consequences of Hg pollution (Supplementary Text 3).

Substantial benefits to the seafood industry from Hg mitigation through MWT

Our study reveals that MWT policies shielded substantial quantities of freshwater seafood from Hg contamination over the past four decades; this safeguarding has prevented substantial economic losses in China's seafood industry. Figure 2a illustrates that the proliferation of MWT plants led to a notable reduction in environmental Hg pollution, reducing its release to rivers by 109 (P10–P90, 104–113; here P10/P90 indicates a 10%/90% probability that the actual results will be below the given values) Mg and directly to adjacent seas by 16.5 (15.7–17.1) Mg in 2022 (Supplementary Data 3). We established a new bioconcentration factor (BCF) database and estimated that this reduction led to a notable increase of 9.2 (8.6–9.7) megatonnes in Hg-safe freshwater seafood production in 2022, aligning with the World Health Organization's (WHO) food safety standard and China's national standard of 0.5 mg kg^{-1} MeHg[38,39] (Fig. 2b). Without MWT, 30% of China's freshwater seafood in 2022 would have exceeded the standards, making it hazardous owing to high Hg levels (Supplementary Data 4 and 5). Using the US Environmental Protection Agency's (EPA) criterion of 0.3 mg kg^{-1} MeHg[40] and the US Food and Drug Administration's "Best choice" value of 0.15 mg kg^{-1} MeHg[41], the violation rates of freshwater seafood would have been 44% and 68%, respectively. Remarkably, these Hg-safe production increases surpassed the annual production of freshwater seafood in India, the world's second-largest producer[17], by a factor of up to three (Supplementary Data 6).

The increase in Hg-safe production due to MWT policies differs notably across freshwater seafood species. As depicted in Fig. 2c, without the MWT industry, 65% of red swamp crayfish (the most popular freshwater invertebrate in China) would have exceeded WHO's and China's safety standards by 2022. Chinese mitten crabs (50%), common carp (41%) and catfish (40%) followed this trend. These high-trophic-level species possess greater Hg bioconcentration abilities and are more sensitive to circumambient Hg levels. In contrast, snakehead fish (1.3%), tilapia (4.4%) and pond mussels (2.0%) experienced smaller increases (Fig. 2d), mainly because of their lower trophic levels. We also observed that the increase in Hg-safe production of fishery capture (41% in 2022) notably exceeded that of aquaculture (29%). This disparity stems from the longer life cycles and exposure of wild seafood to a more diverse environment and diet than their farmed counterparts[36]. Further analyses of Hg-safe production increases in various species across the country are provided in Extended Data Fig. 1 and Supplementary Texts 4 and 5.

From 1980 to 2022, the cumulative increase in Hg-safe freshwater seafood production meeting WHO's and China's safety standards due to the MWT industry reached 146 (137–154) megatonnes, accounting for 20% of the total national freshwater seafood production during this period (Extended Data Fig. 2). Given that Hg-unsafe freshwater seafood is not allowed to circulate in the market[39], the MWT policies in China might circumvent a potential deficit of approximately US$498

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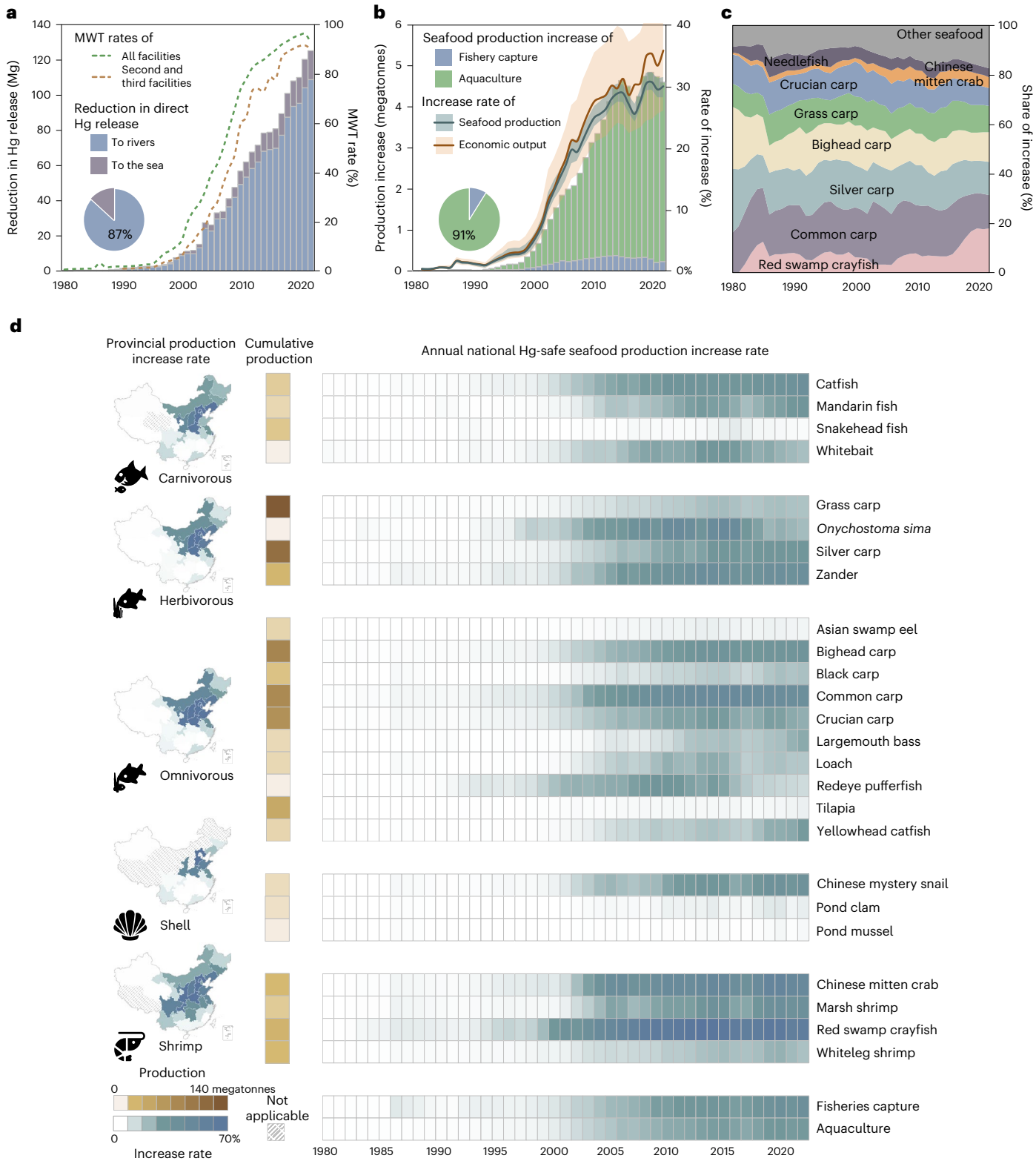


Fig. 2 | Reduction in Hg release contributed by MWT policies and associated Hg-safe freshwater seafood production increases in China during 1980–2022. **a**, Reduction in national Hg release to rivers and directly to the seas. Inset: the proportions of these releases in 2022. **b**, National increases in Hg-safe freshwater seafood production, disaggregated by production activity. Inset: the proportions of these activities in 2022. The dark green line and the brown line show the rates of production increases in Hg-safe freshwater seafood and economic output increases in China's freshwater fishery industry, respectively. **c**, National increase shares of common freshwater seafood species. **d**, National increase rates of Hg-safe seafood production of common freshwater seafood species and production activities. The maps show provincial production increase rates of Hg-safe freshwater seafood in major categories. The confidence of the uncertainty intervals is 80%, as shown by the shaded areas. Base map in **d** from https://www.tianditu.gov.cn/. Icons in **d** adapted from Font Awesome Team at Icon Archive (www.iconarchive.com) under a Creative Commons licence CC BY 4.0.

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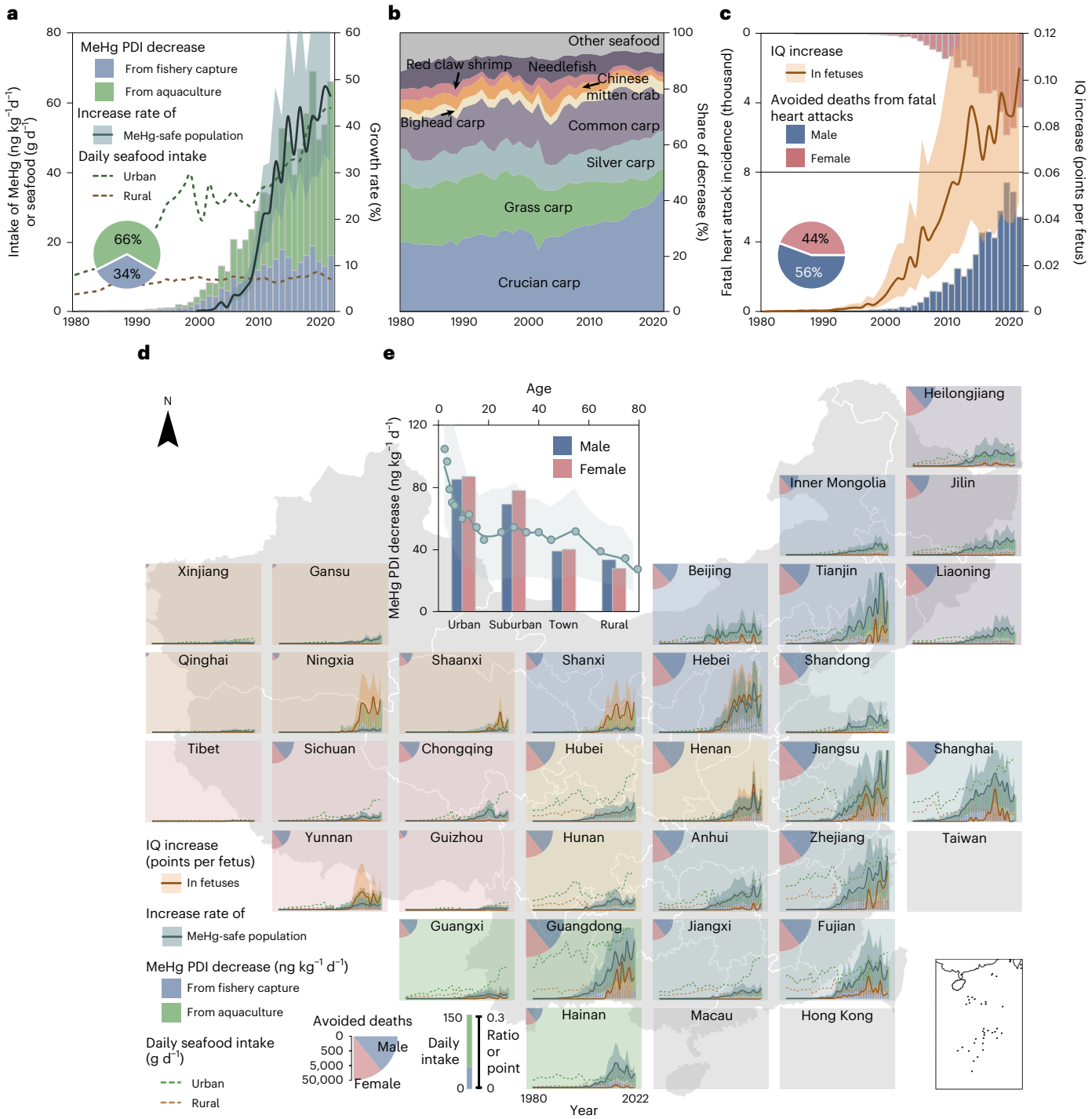


Fig. 3 | Changes in MeHg exposure through freshwater seafood consumption contributed by MWT policies and associated health benefits in China during 1980–2022. a, National decrease in MeHg exposure, disaggregated by production activity. Inset: the proportions of these activities in 2022. The dark green line shows the increase rate of the population exempt from excessive MeHg exposure. The green dashed line and the brown dashed line show the daily freshwater seafood consumption of urban and rural residents, respectively, in China. **b**, Contributions of common freshwater seafood species to national MeHg exposure decreases. **c**, National avoided deaths from fatal heart attacks in males and females. Inset: shows their proportions in 1980–2022. The orange line shows national IQ increases per fetus. **d**, MeHg exposure decreases, disaggregated by production activity. The dark green lines and orange lines remain the same as those in **a** and **c**, respectively. The pie charts show the cumulative provincial avoided deaths from fatal heart attacks disaggregated by gender during 1980–2022. **e**, National decreases in MeHg exposure in 2022, disaggregated by gender and type of residence. The teal dots and line show national decreases in MeHg exposure in 2022, disaggregated by age group. The confidence of the uncertainty intervals is 80%, as shown by the shaded areas. Base map in **d** from https://www.tianditu.gov.cn/. The subplot indicates the South China Sea.

(469–530) billion, equalling 23% of the cumulative freshwater fishery industry's economic output (Supplementary Data 7). Importantly, the increase in production also affected China's export of Hg-safe freshwater products and probably enhanced seafood safety worldwide. For example, China exported 2,900 tonnes of freshwater seafood to the United States in compliance with WHO's food safety standards for MeHg, representing almost 3% of the United States' total freshwater seafood imports. Overall, we suggest that China's MWT policies could alleviate Hg pollution and enhance freshwater seafood safety.

Substantial health benefits from enhanced seafood safety

A central insight from this study is that without the implementation of MWT policies, the rapid increase in China's anthropogenic Hg emissions into the atmosphere and dietary changes over the past four decades could have amplified Hg-related premature deaths by a factor larger than two. We evaluated the impacts of the MWT industry and dietary changes on human health concerning Hg exposure by integrating inter-regional seafood transport matrices, updated consumption inventories and dose–response relationships. We found that probable daily intake (PDI) of MeHg decreased by 50.6 (33.8–145.5) ng kg^{-1} d^{-1} in 2022—equating to 90% of the current MeHg PDI for China's population—solely due to the enhancement in China's freshwater seafood safety related to MWT policies during 1980–2022 (Supplementary Data 8). In contrast, increased freshwater seafood consumption alone during this period led to a 45.7 ng kg^{-1} rise in MeHg PDI. Without MWT facilities, almost 50% of China's population would have exceeded the EPA's Reference Dose for MeHg (100 ng kg^{-1} d^{-1})[42], and 19% would have been at risk according to the Provisional Tolerable Weekly Intake (1,600 ng per kg per week) proposed by the Joint FAO/WHO Expert Committee on Food Additives (Fig. 3a)[43]. Further analyses of the benefits of consuming various Hg-safe species are detailed in Extended Data Fig. 3 and Supplementary Text 6.

Surprisingly, the health benefits derived from MWT-driven Hg mitigation greatly offset the adverse effects of atmospheric Hg emissions. We estimated that reduced MeHg exposure from freshwater seafood due to the MWT industry yielded a gain of 0.10 (0.05–0.21) IQ points per fetus and prevented approximately 9,600 (4,100–18,200) fatal heart attack deaths in 2022 alone (Fig. 3b,c and Supplementary Data 9 and 10). When coupled with changes in diet, we calculated that 102,000 (95,400–113,000) deaths due to heart attacks were avoided nationwide during 1980–2022. These findings display a degree of uncertainty, primarily due to variations in health risk assessment methods. Nonetheless, using the same health risk evaluation and MeHg intake metrics as in this study, previous research estimated that atmospheric Hg emissions caused 5,600–7,400 deaths per year in China in the early 2010s[23,25]. That is, without MWT policies over the past four decades, the annual death toll related to MeHg intake in China could have surpassed 15,000 deaths per year in the past decade.

Our research highlights notable regional and demographic discrepancies in the effects of MWT policies, primarily driven by differences in dietary habits, age distribution and levels of MWT development. Specifically, densely urbanized provinces in East and South China, such as Jiangsu, Zhejiang and Guangdong, reaped substantial health benefits (Fig. 3d). MWT prevented 30,000, 17,000 and 27,000 Hg-related deaths during 1980–2022, respectively, due to the high seafood consumption, increased MWT investment and standards in these regions. As Fig. 3e shows, although urban and suburban residents faced higher MeHg exposure risks, they received more protection from MWT than those in town and rural areas (77–213%). Furthermore, our study indicates that urban residents under 20 years old would have suffered 18–64% higher MeHg exposure without MWT, with adolescents (12–18 years old) in municipalities such as Beijing (65–225% higher MeHg exposure), Tianjin (31–115%) and Shanghai (28–71%) being particularly vulnerable (Extended Data Fig. 4). This highlights the elevated MeHg exposure risk for young urbanites without MWT protection. Additional insights are provided in Supplementary Text 7. These findings underscore the pressing need for regions, particularly those with high seafood consumption and a lack of advanced wastewater treatment, to develop strategies that counteract health risks from MeHg exposure and enhance seafood safety, especially for vulnerable demographics such as children, pregnant women and seniors[44,45].

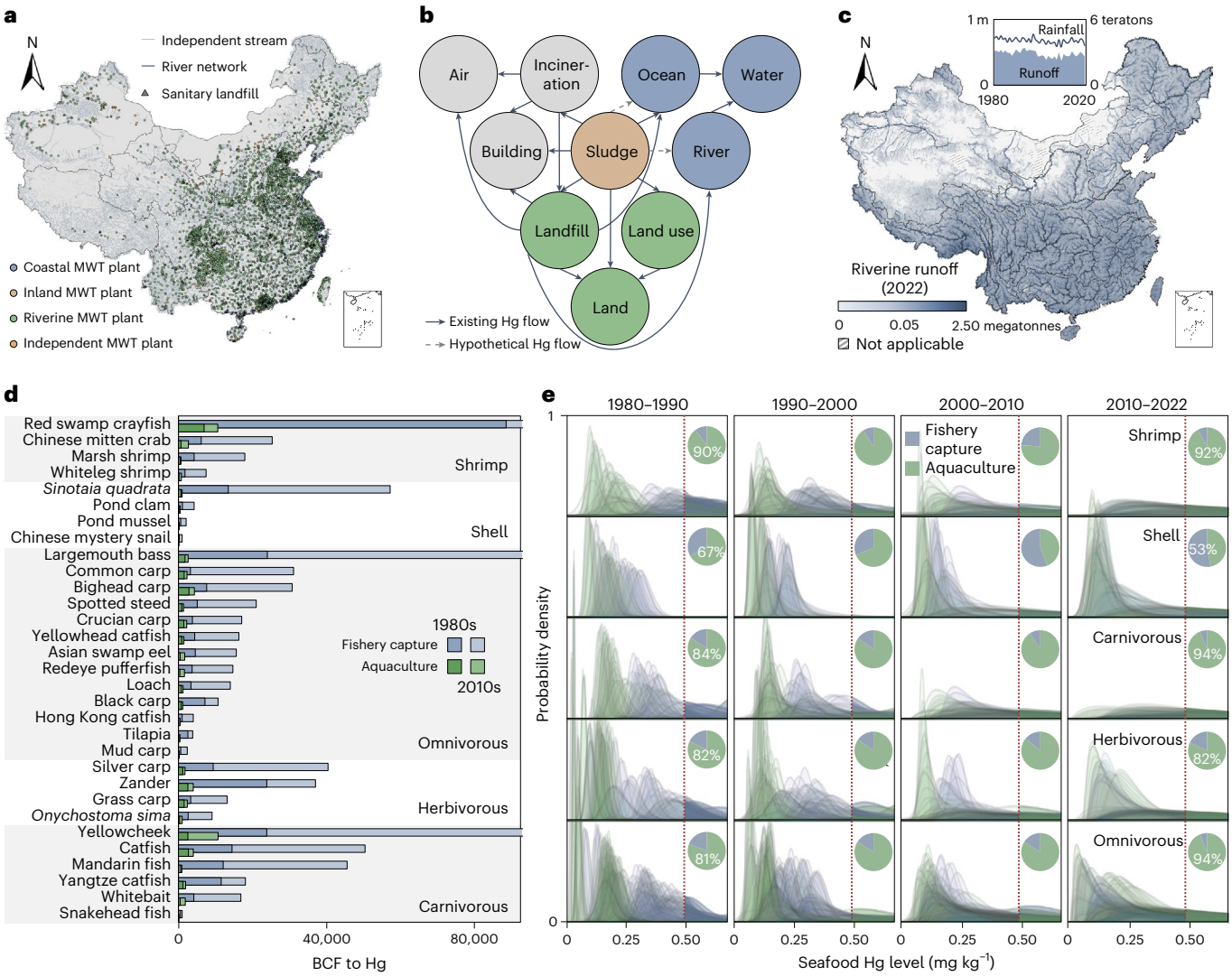
Strategy for counteracting the health impacts of Hg pollution

This study attempted to quantify the impact of China's wastewater policies on seafood safety and Hg exposure risk by developing a biogeochemical model that traces the entire chain of aquatic Hg from its sources to health risks (Fig. 4). The novelty of our study lies in its comprehensive assessment of the impact of China's MWT policies on seafood safety and public health within the global food system. We present a strategy that not only reduces Hg contamination in freshwater seafood but also maximizes the health benefits of increased seafood consumption, which has quadrupled in China over recent decades[46]. We show how MWT policies have counteracted the potential increase in MeHg intake due to growing seafood consumption by notably lowering MeHg exposure through effective wastewater treatment. This dual benefit—enhancing nutritional intake while minimizing health risks—offers a powerful approach to improving food safety and public health, especially in regions heavily affected by aquatic Hg pollution and dependent on freshwater seafood. For instance, the unexpected success of China's MWT policies in improving seafood safety has profound implications for low-to-middle-income countries in Africa, Asia and South America. These countries struggle with severe aquatic Hg pollution and rely heavily on freshwater seafood as a crucial part of their diet[47,48]. Our findings provide a blueprint for global strategies to combat Hg pollution and promote food safety across geopolitical boundaries.

As the leading global Hg emitter and seafood provider, China still confronts an urgent necessity to further mitigate anthropogenic Hg pollution[16,49,50]. Violations of permissible levels of conventional pollutants such as organic matter, nitrogen and phosphorus also persist in tributaries and small streams nationwide, underscoring the imperative for more robust pollution mitigation policies. These policies present opportunities to optimize wastewater treatment and enhance Hg removal efficiencies across the country. More broadly, municipal wastewater is an important source of many critical aquatic pollutants, including antibiotics, polychlorinated biphenyls and perfluorooctane sulfonate/acid, which, like Hg, can bioaccumulate in seafood[51–53]. We anticipate that further enhancements to China's MWT policies will continue to decrease

Fig. 4 | Study methodology. a, The compilation of Hg loading inventories of sewage sludge generated by China's MWT plants. **b**, Unit-based dynamic material flow analysis of Hg loadings in sewage sludge. The solid arrows and the dashed arrows denote actual flows and hypothetical flows if MWT was absent, respectively. **c**, The national river network model established in this study on the basis of level-12 watersheds. **d**, BCFs of common freshwater seafood species, disaggregated by production activity and decade. These data were employed for bioconcentration simulation in this study. **e**, The ecological risk assessment of common freshwater seafood under WHO's and China's national standards of MeHg (0.5 mg kg^{-1}), disaggregated by major species category and production activity. The vertical axes represent the probability density of seafood Hg levels, ranging from 0 to 1. The dark areas under the curves indicate the probability of seafood Hg levels exceeding WHO's and China's standards. Insets: the pie charts show the proportions of risky freshwater seafood from different production activities. **f**, Left: interprovincial trade of raw and refrigerated freshwater seafood, the processing of freshwater seafood and interprovincial trade of processed seafood products in 2020. Right: the annual per capita consumption rates (kg) of freshwater seafood for urban and rural residents in each province. **g**, The settings of selected parameters of scenarios in 2022 against those in 1980 (baseline). The vertical axes represent the relative scales of the parameters in diverse scenarios compared with those in baseline scenario. Base map in **a** and **c** from https://www.tianditu.gov.cn/.

https://doi.org/10.1038/s43016-024-01093-9



these pollutants and improve seafood safety. However, current MWT policies in China predominantly rely on national command-and-control environmental regulations. Further enhancing national standards for MWT effluent will probably require substantial investments in advanced facilities. Given the geographical diversity in wastewater properties, environmental conditions and economic development levels, uniform MWT policies have led to various deficiencies in the MWT system, including wasteful investment costs, and need to be improved.

We recommend that China adopt region-differentiated strategies for enhancing its MWT systems on the basis of comprehensive cost–benefit analyses. Specific strategies might include the following

Southern regions. Regular pipeline maintenance could offer greater cost-effective benefits in regions with high rainfall and inflow infiltration. Repairing damaged pipelines can reduce the operational burden on MWT systems and improve the efficiency of pollution treatment by facilitating anaerobic sludge digestion and lowering operational and energy costs.

Large northern cities (for example, Beijing). In areas with dense populations and arid climates, advanced treatment technologies can more effectively mitigate local aquatic pollution and enhance seafood safety. Additionally, these technologies increase the availability of reclaimed and recycled water, thus alleviating local water scarcity.

Provinces with low wastewater collection (for example, Hubei, Guangxi and Tibet). It is of utmost importance to prioritize expanding pipeline construction to enhance wastewater collection and address relatively high pollutant releases from untreated municipal wastewater in areas with low wastewater collection (Supplementary Fig. 1).

Provinces with suboptimal treatment rates (for example, Liaoning, Jiangxi and Xinjiang). Retrofitting to expand MWT capacity can be a straightforward and effective mitigation measure for improving Hg removal efficiency and reducing local water pollution in areas with modest treatment rates.

Disparate regional seafood production and consumption. China exhibits notable regional disparities in producing and consuming freshwater seafood (Fig. 4f). Implementing stricter effluent standards in primary production regions can yield substantial seafood safety benefits at relatively modest costs, thereby alleviating MeHg exposure health risk. Additionally, creating interregional transfer payment mechanisms can boost investments in MWT facilities in higher-fishery-production areas.

Potential multi-media risks from disposal. While MWT averts aquatic Hg pollution, it transfers Hg into sewage sludge, whose disposal, such as land application and incineration, may engender further health risks. Despite China's policy banning the use of sludge in fertilizer production[54], the recent prevalence of sludge incineration might intensify atmospheric Hg pollution and exposure[55]. More research is needed to precisely elucidate these multi-media risk issues.

Integrated industrial wastewater policies. The potential for reducing aquatic pollutants may also originate from industrial sectors. For instance, non-ferrous metal smelting and vinyl chloride monomer production already account for about 40% of total aquatic Hg releases. Despite the complexity of treatment processes (Supplementary Text 8), integrating industrial wastewater policies with MWT policies is a valuable consideration.

Overall, it is crucial to adapt MWT policies to regional conditions. By addressing the distinct challenges and opportunities each region faces, these strategies can contribute to the development of a more extensive, efficient and resilient national water system, ultimately enhancing seafood safety and human health across China.

Global concerns regarding Hg contamination of seafood are increasing[13,16]. Our study presents a comprehensive assessment of the impacts of Hg, including the important role of wastewater Hg release, a source generally omitted from previous large-scale evaluations. Stepwise model validation and uncertainty analysis (Supplementary Texts 9 and 10) were performed to ensure the robustness of our model. However, due to the model's limitations (Supplementary Text 11) and the absence of wastewater Hg monitoring in many countries[22,56], our model may not fully capture global efforts to reduce Hg risks. Nevertheless, our findings can shape future research and policy. For example, countries need to strengthen the monitoring of wastewater Hg release, incorporate the release into policymaking and prioritize wastewater treatment[57]. Finally, this study does not undermine the importance of reducing atmospheric Hg emissions, which is a fundamental solution for addressing environmental Hg pollution[58–60]. Instead, we highlight a strategy that swiftly mitigates the health impacts of atmospheric Hg emissions in the short term and counteracts them substantially in the long term.

Limitations and perspectives

Our integrated model has certain limitations that warrant further refinement and exploration. The intricate processes by which anthropogenic Hg enters the environment and impacts seafood safety and human health are influenced by numerous intertwined factors. Given this, the sub-models used to describe Hg dynamics had to simplify these processes, resulting in a more basic representation of Hg fate. In addition, our model's focus on the impacts of MWT on Hg in aquatic ecosystems did not account for multi-media health risks from secondary Hg emissions into the atmosphere and soil, or from other food categories. Specifically, the influence of imported seafood, accounting for ~2% of total supply, was not detailed by region and species. Moreover, our model primarily focused on wastewater treatment associated with municipal and industrial initiatives, as rural wastewater and sludge, typically used in farmyards in China, are expected to have limited impact on Hg releases into surface water and seafood[61]. These limitations present opportunities for further improvement, including a better understanding of Hg behaviour and the development of an enhanced integrated model that encompasses a broader range of environmental media and exposure pathways. Such advancements would provide a more comprehensive assessment and inform improved management strategies to minimize Hg-related health risks. For more discussion on limitations and implications for future research, please refer to Supplementary Text 11.

Methods

Analytical framework

In this study, a new comprehensive process-based model was established to investigate the full chain of Hg from its aquatic releases to its impact on human health. As depicted in Fig. 4, the model encompasses eight major components: plant-level inventories of Hg in sewage sludge, unit-based dynamic material flow analysis, a nationwide river network model, a bioconcentration simulation, an ecological risk assessment, an interregional transport model, Hg exposure evaluation and a health risk and impact assessment. The first five components quantify the enhancements in China's freshwater seafood safety caused by MWT. The decrease in Hg levels in freshwater seafood was incorporated into the final three components to evaluate the resulting health benefits. To identify the individual contributions of influential factors, such as MWT, industrial wastewater treatment, aquaculture expansion, overfishing, climate change and others, we conducted scenario simulations related to Hg levels in freshwater seafood (Supplementary Fig. 2).

Hg inventories of municipal sewage sludge

In contemporary MWT plants, Hg is removed from wastewater and relocated to the sewage sludge[62–64]. To gain a precise understanding of

the reduction in Hg release from MWT, we established a new plant-level activity dataset of China's MWT plants, covering 6,132 facilities nationwide from 1980 to 2022, as shown in Fig. 4a. This dataset includes information on the geographical location, operation, treatment capacity, adopted treatment process, treatment water properties, underlying drainage system, common pollutant discharge, discharge pathway and sewage sludge properties of each plant. These data were sourced from official datasets of the Ministry of Housing and Urban-Rural Development of China[65]. Missing data on wastewater discharge and sludge yield were estimated on the basis of annual provincial values from official yearbooks, design capacity and operation year of MWT plants. Transcriptional errors were corrected and recalibrated using official provincial data[55,66,67].

Moreover, a new dataset of 714 plant-level sewage sludge samples with detailed Hg concentrations was assembled through an extensive literature review (Supplementary Table 1). Using datasets of plant-level activity and Hg concentrations, we applied a random forest model to generate Hg concentrations in the sewage sludge for each MWT facility. The random forest model is a well-established machine learning algorithm widely applied in geographical and environmental research[68]. Details on our model's settings and validation are presented in Supplementary Texts 9 and 12. We calculated the plant-level Hg inventories for municipal sewage sludge in China using equation (1):

$$\text{Loading}_{p,t} = C_{\text{sludge},p,t} \times M_{p,t} \quad (1)$$

where p denotes different plants, t denotes the calculation year, Loading denotes the Hg loadings (Mg), C_{sludge} denotes the Hg concentration (μg g^{-1}) of sewage sludge and M denotes sewage sludge production (megatonnes). This study incorporated all the parameters into the (joint) probability density distributions through a Monte Carlo simulation.

Dynamic material flow analysis

In wastewater treatment facilities, Hg contained in sewage sludge can leach into the adjacent environmental media[69]. This unintentional release serves as an important secondary source and impairs efforts to reduce Hg levels in aquatic ecosystems via wastewater treatment[22]. We therefore established a new dynamic material flow model to discern the comprehensive Hg trajectory associated with sewage sludge. As shown in Fig. 4b, Hg flows were sourced from sewage sludge, continued with disposal processes and pathways (that is, landfill, land use, incineration, river and ocean dumping), and ended with release into various sinks (that is, aquatic environments, land, atmosphere and building materials). The analysis was based on the principle of the mass balance of Hg, ensuring that quantities originating from the sources matched those allocated to the sinks[70]. Material flow analysis was performed using equations (2) and (3):

$$\text{Storage}_{p,t,s} = \rho_{s,t} \times \text{Loading}_{p,t} + \sum_{s' \neq s} \rho_{s',t} \times \zeta_{s',s} \times \text{Loading}_{p,t} \quad (2)$$

$$\text{Sink}_{p,t,r} = \rho_{r,t} \times \text{Loading}_{p,t} + \sum_{t' \leq t} \sum_{s} \zeta_{s,r} \times \text{Storage}_{p,t',s} \quad (3)$$

where Loading, Storage and Sink denote the Hg loadings (Mg) of sewage sludge, disposal processes and pathways, and sinks, respectively; s denotes different disposal processes and pathways of sewage sludge; r denotes different sinks; and ρ and ζ denote the proportions (%) of Hg transport/release from the source into a storage location or sink, and from a storage location to another storage location or sink, respectively. Hg accumulation in landfills was considered in the modelling.

The net Hg reduction in a MWT plant discharged into freshwater was calculated using equation (4):

$$R_{b,t} = \sum_{p} \theta_{p,b} \times \left(\text{Loading}_{p,t} - \text{Storage}_{p,t,\text{river}}\right) \quad (4)$$

where R denotes net Hg reduction (Mg), b denotes different sub-watersheds and θ is a binary variable depending on the spatial relationship between plants and watersheds (that is, $\theta_{p,b} = 1$ when plant p located in watershed b; otherwise, $\theta_{p,b} = 0$). The details of the parameters and settings of the model are presented in Supplementary Text 13.

River network model

In recent decades, detailed studies and simulations using physical mechanism-based models have explored the outcomes of aquatic Hg pollution across diverse landscapes. Notable examples include the Soil and Water Analysis Tool[71,72], the Geo-referenced Regional Exposure Assessment Tool for European Rivers model[73], and the Grid-Based Hg Model[74,75]. Studies have validated the efficacy of models for quantitatively addressing both point-source and non-point-source pollution[76,77]. Traditionally, MWT plants have been considered point sources[73]. Using the Markov chain approach, we established a nationwide river network model based on the Soil and Water Analysis Tool stream module[78]. More detailed descriptions of the model and its reliability are provided in Supplementary Texts 9 and 14.

Data on river reaches and level-12 sub-watersheds were sourced from the HydroATLAS database[79,80] and recalibrated using precipitation and surface runoff information extracted from the official water resource bulletins of the provinces and major river basin regions in China (Supplementary Data 11 and 12). Additionally, we integrated a high-resolution precipitation dataset (1 km) using ANUSPLIN v.4.4, which was supported by monitoring inputs from over 2,400 meteorological stations nationwide[81]. As illustrated in Fig. 4c, the longest branch integer in this river network was 526, and the level-12 sub-watersheds covered an average area of 126 km^2, providing our river network model with both precision and reliability. We thus calculated the changes in aquatic Hg concentration ($\Delta C_{\text{aqua,Hg},b,t}$, μg l^{-1}) within each watershed (Extended Data Fig. 5).

Bioconcentration simulation

The most comprehensive approach for linking aquatic Hg loads with Hg in fish tissues involves the mechanistic modelling of Hg circulation and bioaccumulation, exemplified by tools such as Bioaccumulation and Aquatic System Simulator[82–84]. However, these hydrostatic models require site-specific parameterization and calibration to simulate the biotic and abiotic factors controlling bioaccumulation, rendering them inappropriate for nationwide assessments. The BCF is the ratio of the chemical concentration in an organism or biota to the concentration in water[83,85]. It is the most prevalent method for connecting Hg levels in the water column and those in biota tissue and is universally applicable, except in highly contaminated sites[86].

In this study, we compiled a new BCF database for freshwater seafood in China on the basis of an extensive literature review (Fig. 4d). The database encompassed 415 samples from 31 commonly consumed freshwater species in China, representing approximately 95% of the national production in 2022. Details on these species are presented in Supplementary Table 2. The average values for omnivorous species were used for other seafood. The production activities (that is, fishery capture and aquaculture) for the same species were distinguished because of notable disparities in their bioaccumulation characteristics.

We further explored the temporal trends of BCFs for frequently consumed freshwater species using the Mann–Kendall trend test, a non-parametric method designed to identify trends in time-series datasets. As illustrated in Supplementary Fig. 3, BCFs for seafood sourced from fishery capture typically exhibited statistically significant declining trends over the past two decades ($P < 0.1$), whereas those from aquaculture presented no definitive pattern. Moreover, we adopted the average Hg levels spanning the lifetime of seafood from different

production activities to represent the impact of changes in body weight and length, given increasing bioaccumulation with growth[35]. To align with the consumption rates and interregional food trade data, the Hg levels in the watersheds were merged into provincial values. A simplified assumption posits uniform seafood production densities across surface runoff in small sub-watersheds. The Hg levels in seafood at the provincial scale were subsequently determined using weighted average levels. Watersheds across several provinces were partitioned according to their individual areas. We used the optimal correlation between MeHg and Hg levels, as determined in our previous study[36], to simulate the MeHg levels[87]. The changes in Hg levels in freshwater seafood species were calculated using equations (5) and (6):

$$\Delta C_{\mathrm{biota,Hg},f,s,i,t} = \frac{1}{V_{i,t}}\left(\sum_{b\in i}\frac{V_{b,t}}{T_s}\sum_{t'=t-T_s}^{t}\mathrm{BCF}_{f,s,t'}\times\Delta C_{\mathrm{aqua,Hg},b,t'}\right) \quad (5)$$

$$\Delta C_{\mathrm{biota,MeHg},f,s,i,t} = \beta_s \times \Delta C_{\mathrm{biota,Hg},f,s,i,t} + \alpha_s \quad (6)$$

where ΔC_{aqua} and $\Delta C_{\mathrm{biota}}$ denote the changes of Hg levels in water and seafood (μg kg^{-1}), respectively, f denotes different freshwater species (Fig. 4d), s denotes different production activities, T denotes the average growth cycle (year), i denotes different provinces, V denotes surface runoff (gigatons), and β and α denote parameters standing for the relationship between MeHg and Hg levels.

Ecological risk assessment

An ecological risk assessment is a systematic method used to evaluate the likelihood of a particular event occurring under certain circumstances[88]. Its purpose is to provide a quantitative foundation for balancing and comparing the risks associated with environmental issues, thereby improving the estimation and understanding of these risks[89]. In this study, we analysed the ecological risk to quantify the production increase of Hg-safe freshwater seafood. The current Hg levels in seafood were fitted as log-normal distributions drawn from data sourced from the literature[28,90]. We employed the Monte Carlo method to generate probabilistic distributions of changes in Hg levels in seafood. Further details on the parameters and settings used are provided in Supplementary Text 10. As shown in Fig. 4e, 1,922 probabilistic distributions were generated across species, production activity and province for each year.

The cumulative probabilities above the specific threshold describe the risks of China's seafood Hg levels surpassing specific food safety standards for Hg in the absence of MWT. Broadly speaking, this represents a percentage increase in Hg-safe seafood production attributable to China's MWT efforts. We calculated the cumulative probabilities of seafood MeHg levels exceeding the EPA's food safety standard (0.3 mg kg^{-1})[40], the US Food and Drug Administration's "Best choice" screening value (0.15 mg kg^{-1})[41] and the WHO's reference level as well as China's national standard (0.5 mg kg^{-1})[38,39]. We collected seafood production data for each freshwater species from the official yearbooks across all provinces (Supplementary Data 4). Missing data were estimated on the basis of freshwater seafood production disaggregated by activities during 1980–1998 and by species in 1999. The changes in the production of seafood that met certain food safety standards were calculated using equation (7):

$$\Delta P_{f,s,i,t} = P_{f,s,i,t}\times\left(\int_0^{l_k}\left(\Delta C_{\mathrm{biota,MeHg},f,s,i,t} + C_{\mathrm{biota,MeHg},f,s,i,t}\right)(x)\,\mathrm{d}x - \int_0^{l_k} C_{\mathrm{biota,MeHg},f,s,i,t}\,(x)\,\mathrm{d}x\right) \quad (7)$$

where P and ΔP denote freshwater seafood production (megatonnes) and its changes in the absence of the MWT industry (megatonnes), respectively; the x denotes the random variable representing the probability distribution of Hg levels in seafood (μg kg^{-1}); and k and l denote the index and criterion value of specific food safety standards for Hg, respectively.

Market price data for seafood were gathered from the National Bureau of Statistics of China to evaluate the increase in economic value (Supplementary Data 6). Missing data were estimated using producer price indexes of freshwater products from aquaculture and fishery capture. The economic outputs were adjusted to the currency value of 2022 for consistency, according to China's consumer price index from World Bank Open Data[91]. The data prior to 1986 were imputed using linear interpolation. The average exchange rate of the US dollar to the Chinese yuan in 2022 (6.7261) was employed. For context, we extracted the gross value output of the national fishery industry from official yearbooks[92,93]. To assess the export growth of each partner nation, we used the global fish trade dataset from the Food and Agriculture Organization statistics database (FAOSTAT)[17], integrated with the production growth rates of each freshwater species in China.

Interregional transport model

Given the disparity between seafood production and consumption regions in China and the existing domestic food transportation network, we incorporated interregional seafood trade into our analysis. In previous studies, China's provincial monetary multiregional input–output tables have been used to estimate the transportation of goods among provinces. However, this approach often faces problems owing to its inadequacy in accounting for passers-by goods[58]. Drawing on the quasi-input–output model used in electricity transmission analyses[94,95] and the water-filling algorithm implemented in our previous research[58], we developed an algorithm to derive a physical transport matrix from the multiregional input–output table to resolve the biases. We constructed a new interprovincial transport matrix for each species in each year (Extended Data Figs. 6 and 7). The details of the methodology, data and settings are provided in Supplementary Text 15.

Hg exposure

We first simulated the changes in MeHg levels in the seafood consumed in each province. Using the interprovincial transport matrices, we converted the changes in MeHg levels of freshwater seafood species produced in each province i into those of seafood consumed ($\Delta C_{\mathrm{consumption}}$, μg kg^{-1}) in each province j, as shown in equation (8):

$$\Delta C_{\mathrm{consumption,MeHg},f,s,j,t} = \frac{\sum_i \Delta C_{\mathrm{biota,MeHg},f,s,i,t}\times r_{i,j,f,t}}{\sum_i r_{i,j,f,t}} \quad (8)$$

The same transport matrix was applied to seafood of a given species from both fishery capture and aquaculture because Chinese consumers hardly differentiate freshwater seafood from different production activities[96]. Next, we compiled a provincial inventory of seafood consumption by the inhabitants. Compared with previous inventories of seafood consumption[23,44,97], our annual provincial inventory further breaks down into data for each species, production activity, gender, age group and more detailed type of residence (urban, suburban, town and rural areas; Fig. 3e). This helped us identify potential health risks for vulnerable demographics (for example, children, young urbanites and pregnant women)[98]. Since imported freshwater seafood is not affected by the domestic MWT industry, corresponding consumption was excluded. The details regarding the establishment of the consumption inventory are provided in Supplementary Text 16. The changes in the PDI of MeHg were calculated to evaluate the impact of MWT on human MeHg exposure using equation (9):

$$\Delta \mathrm{PDI}_{\mathrm{MeHg},j,a,g,c,t} = \sum_f\sum_s \frac{\Delta C_{\mathrm{consumption,MeHg},f,s,j,t}\times \mathrm{Consumption}_{f,s,j,a,g,c,t}}{\mathrm{BW}_{j,a,g,c}} \quad (9)$$

where ΔPDI denotes the change in PDI of MeHg (μg kg^{-1} d^{-1}), Consumption denotes seafood consumption per capita (kg capita^{-1} yr^{-1}) and BW denotes the body weight (kg) of the Chinese population within age group a, gender g and type of residence c. We used population-weighted averages for the national and regional estimations. Population data were obtained from official yearbooks[46]. Data detailing the body weight of the Chinese population disaggregated by age group, gender and type of residence were collected from the Exposure Factors Handbook of Chinese Population for Children[99,100] and Adults[101]. The impact of change in freshwater seafood consumption was analysed, assuming national consumption levels persisted since 1980.

Health risk and impact

The detrimental health effects associated with dietary MeHg exposure include neurodevelopmental and cardiovascular effects[8]. In particular, prenatal exposure to MeHg in maternal bodies can lead to IQ decreases in the fetus and persistent neurodevelopmental delays[7,9,12]. While a correlation between MeHg intake and cardiovascular consequences has been posited for over two decades, considerable uncertainty remains owing to inconsistent findings[102]. Nonetheless, recent research has suggested that MeHg can counteract the cardioprotective effects of omega-3 polyunsaturated fatty acids[103]. Instead of a direct cause, dietary MeHg exposure is perceived as a synergistic element that accelerates fatal heart attacks. Despite these incongruities, fatal heart attacks have been acknowledged as a critical MeHg-related health impact on the basis of the EPA's recommendations and other studies[9,104].

Following the published literature[23,97,105], we applied the linear dose–response relationships between MeHg intake and associated IQ decreases and the log-linear relationship between MeHg intake and fatal heart attacks, which were adjusted to conduct a counterfactual prediction in this study (equations (10)–(12))[106]:

$$\Delta IQ_{j,c,t} = \sum_{a\geq 18}^{35} \varepsilon \times \lambda_{\text{Female}} \times \xi \times \Delta PDI_{\text{MeHg},j,a,\text{Female},c,t} \times BW_{j,a,\text{Female},c} \quad (10)$$

$$\chi_{j,a,g,c,t} = \omega \times (1 - e^{-\epsilon \times \lambda_g \times \xi \times \Delta PDI_{\text{MeHg},j,a,g,c,t-\tau} \times BW_{j,a,g,c}}) \quad (11)$$

$$\Delta CF_{j,g,c,t} = \sum_{a\geq 30} \frac{\chi_{j,a,g,c,t}}{1-\chi_{j,a,g,c,t}} \times \kappa_{j,a,g,c,t} \times Popu_{j,a,g,c,t} \quad (12)$$

where ΔIQ and ΔCF denote the changes in IQ (points) and deaths due to fatal heart attacks in the absence of MWT, respectively; ξ, λ and ε are the conversion coefficients of MeHg intake to blood (d l^{-1}), blood to hair (l g^{-1}) and hair to IQ (points per (μg g^{-1})), respectively; ε denotes the heart attack–hair MeHg coefficient, which reflects the relationship between hair MeHg levels and fatal heart attack risks (g μg^{-1}); ω denotes the probability that reflects the causality of the association between the hair MeHg level and heart attack risk, which is a subjective probability of one third for the causal epidemiological associations (that is, $\omega = 1$) and two thirds for no causal associations (that is, $\omega = 0$); κ denotes the current incidence risks of fatal heart attacks (%); χ denotes the adjustment factor for counterfactual prediction; and Popu denotes number of people.

For IQ, females aged between 18 and 35 years were considered to be of reproductive age. People over 30 years of age are regarded as vulnerable to cardiovascular disease. The delay in heart attack death was characterized by the parameter τ (years), which aimed to determine the duration between changes in MeHg consumption and subsequent fatal heart attacks. The central tendency was estimated as six years according to previous studies[97,105]. In addition, we conducted a health risk assessment to determine the number of individuals at risk of excessive MeHg exposure in the absence of MWT. The details of the health risk assessment methodology are provided in Supplementary Text 17.

Scenario analysis

On the basis of the full-chain biogeochemical model, we simulated six hypothetical scenarios: Municipal, Industrial, Aquaculture-constant, Capture-constant, Climate-neutral and Environment-neutral. These scenarios were designed to emulate the impacts of diverse influential factors as highlighted by the hypotheses addressing the contradiction between increasing atmospheric Hg emissions and decreasing Hg levels in freshwater seafood in China (Fig. 4f)[28,35,107,108]. Under the Municipal scenario, we assumed that no new MWT plants have been constructed since 1980. Given that only a few modern MWT facilities existed in China in 1980 (ref. 65), this scenario elucidates the impact of the developments in the MWT industry. In the Industrial scenario, we posited that no new industrial wastewater treatment installations were established post-1980 in China, thereby shedding light on the benefits of industrial wastewater treatment development. We devised the Aquaculture-constant and Capture-constant scenarios to determine the impacts of massive aquaculture practices and aggressive fishery capture, respectively. In these scenarios, the corresponding production, BCFs and seafood consumption levels were consistent with 1980 levels. Additionally, we set up the Climate-neutral and Environment-neutral scenarios to model the impacts of climate change and other external factors (that is, atmospheric deposition, surface sinkage, soil erosion and fugitive releases), respectively. Further details on the scenario descriptions and settings are presented in Supplementary Table 3.

Model validation

Our integrated model comprises several sub-models, including plant-level inventories based on the random forest model, the dynamic material flow model, the river network model, the bioconcentration model, the ecological risk model, the interregional transport model, the Hg exposure model and the health impact model. We conducted a stepwise validation of each sub-model based on diverse observations to bolster the robustness and credibility of our findings. Detailed validation approaches and results are presented in Supplementary Text 9.

Uncertainty analysis

A Monte Carlo simulation was conducted 10,000 times to evaluate the overall uncertainties of the factors mentioned above in accordance with previous studies[109]. Additionally, two bootstrap simulations, each executed 10,000 times, were employed to assess the uncertainties associated with increases in Hg-safe seafood production and risk-free population. An 80% confidence level was obtained, represented by a P10–P90 confidence interval. For the primary uncertainty parameter settings and further discussion, please refer to the preceding section and Supplementary Text 10.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The underlying data used in this study are provided in the Supplementary Information or are available from the sources listed in the main text and the Supplementary Information. Source data are provided with this paper.

Code availability

Data collection and analyses were conducted in Microsoft Excel 2020, Python v.3.9.10, ArcGIS v.10.8, ANUSPLIN v.4.4 and WebPlotDigitizer v.4.7. The code is available from the corresponding authors upon reasonable request.

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Acknowledgements

This work was partially funded by the National Natural Science Foundation of China (grant nos 41821005 and 42476127). M.L. is supported by the Fundamental Research Funds for the Central Universities (grant no. 7100604309) and the Laboratory for Earth Surface Processes, Ministry of Education, Peking University. Q.Z. acknowledges support from the China Postdoctoral Science Foundation (grant no. 2022M720005) and the Beijing Natural Science Foundation (grant no. 8244068). X.C. acknowledges support from Peking University-BHP Carbon and Climate Wei-Ming PhD Scholars (grant no. WM202209) and the High-Performance Computing Platform of Peking University.

Author contributions

X.C., M.Y., J.M. and X.W. designed the research. X.W. and M.L. acquired the funding needed to complete the study. M.Y. and X.C. performed the data collection. X.C. conducted the data processing and modelling in close discussion with H.Z. and S.M. X.C., M.Y. and M.L. wrote the original paper in close discussion with C.Y., J.M. and X.W. Y.C., Q.Z. and X.D. provided important data to help complete the work. All authors contributed to paper revision and completion.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s43016-024-01093-9.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43016-024-01093-9.

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Peer review information *Nature Food* thanks the anonymous reviewers for their contribution to the peer review of this work.

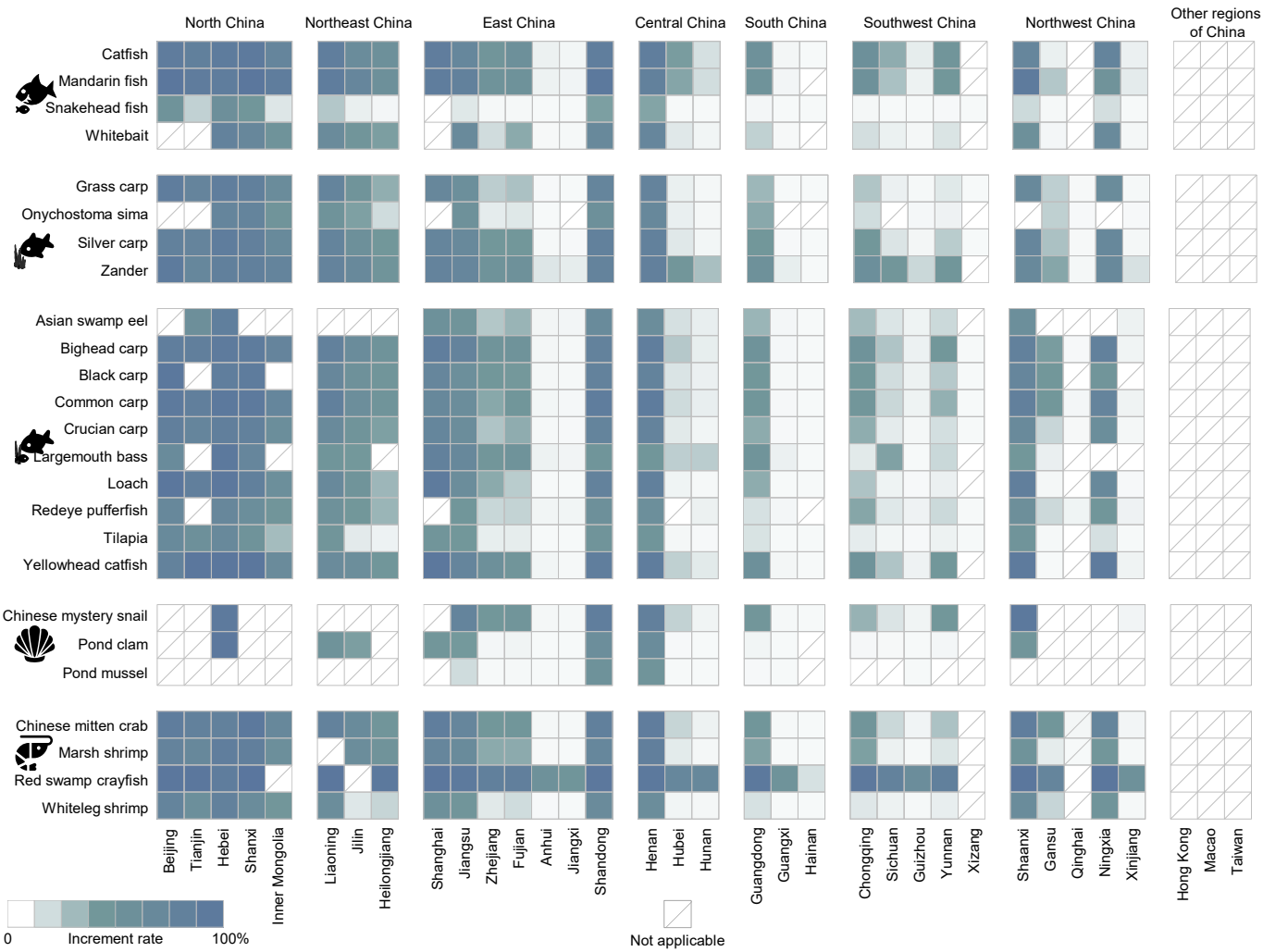
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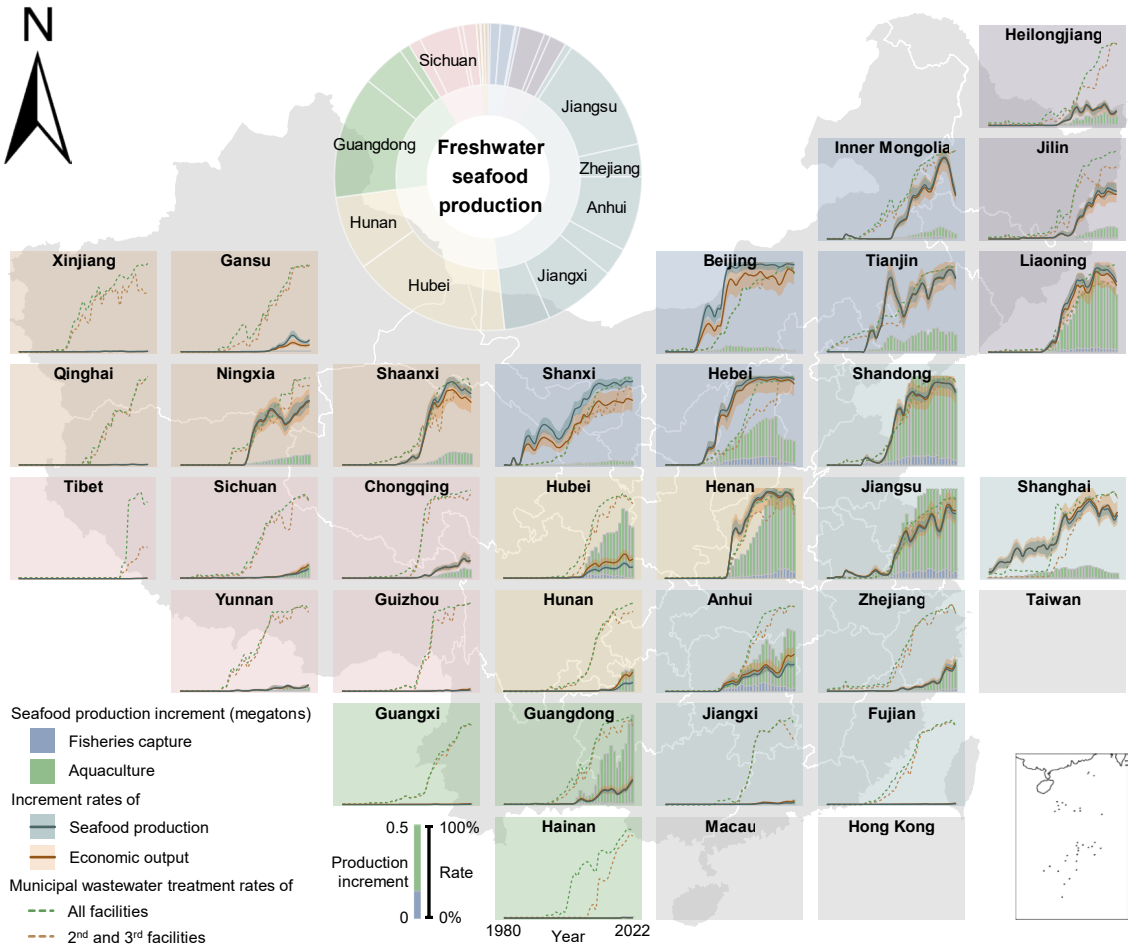
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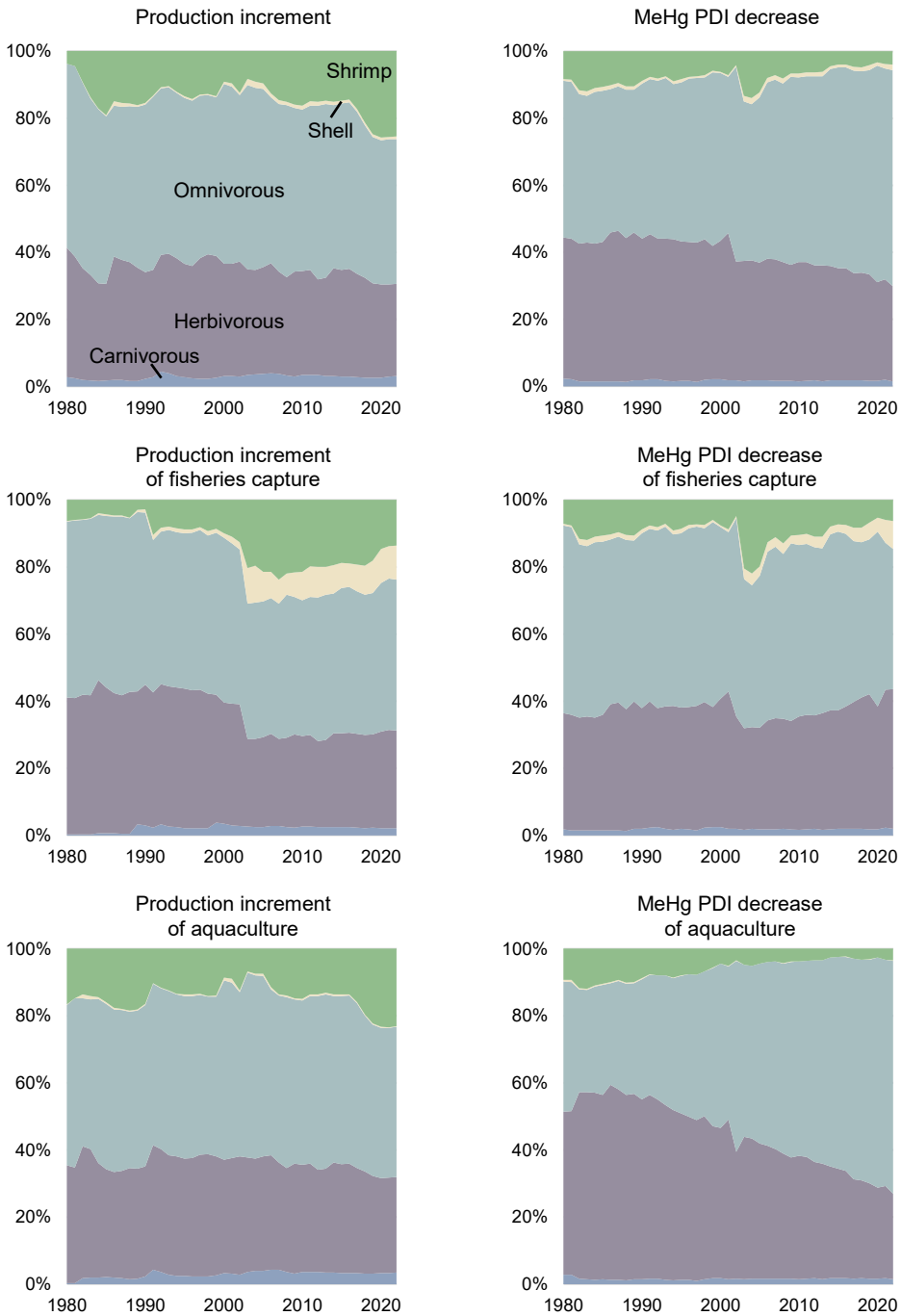
Extended Data Fig. 1 | Hg-safe seafood production increases rates in different provinces in 2022. The shades of the color blocks represent the rates of production increases contributed by MWT for each species across different provinces.

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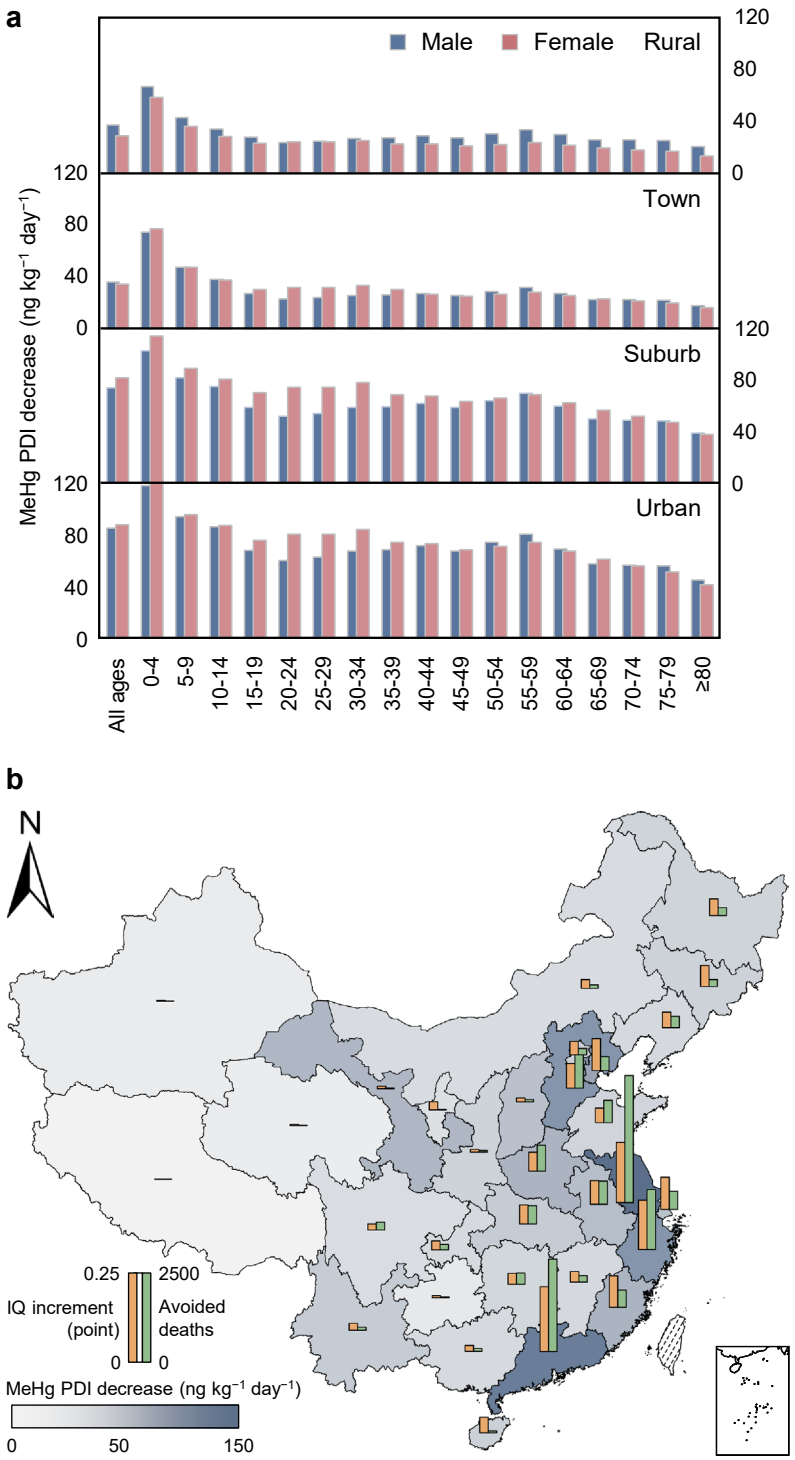
Extended Data Fig. 2 | Provincial Hg-safe freshwater seafood production increases, disaggregated by production activity. The pie plot shows provincial seafood production in 2022. The dark green line and brown line show the rates of production increases in Hg-safe freshwater seafood and economic output increases in China's freshwater fishery industry, respectively. The confidence of uncertainty intervals is 80%, as shown by the shaded areas. Base map from https://www.tianditu.gov.cn/.

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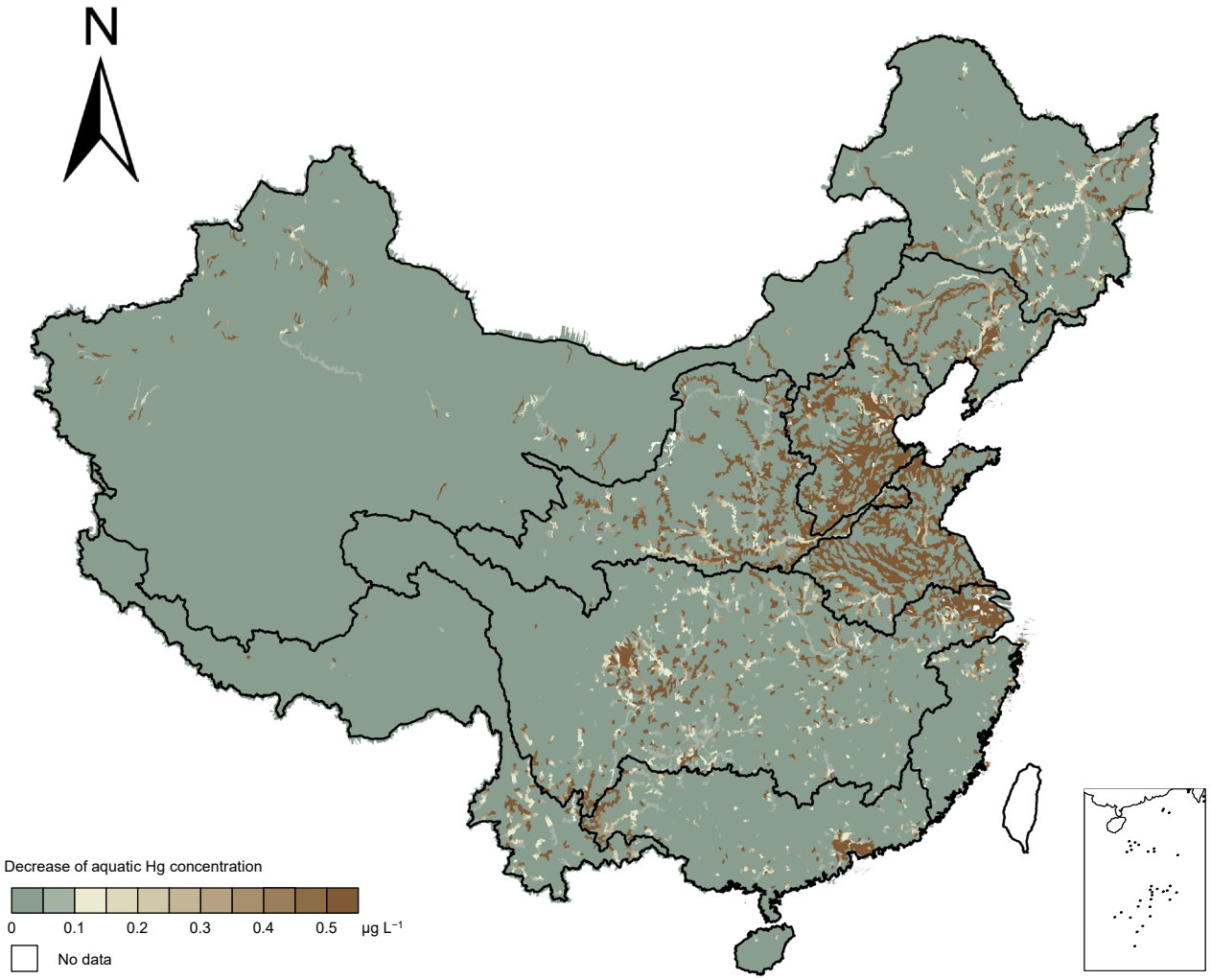
Extended Data Fig. 3 | Changes in Hg-safe seafood production and contribution shares to PDI of MeHg. National increase shares of Hg-safe seafood production and contribution shares to MeHg PDI decreases by major category of freshwater seafood species.

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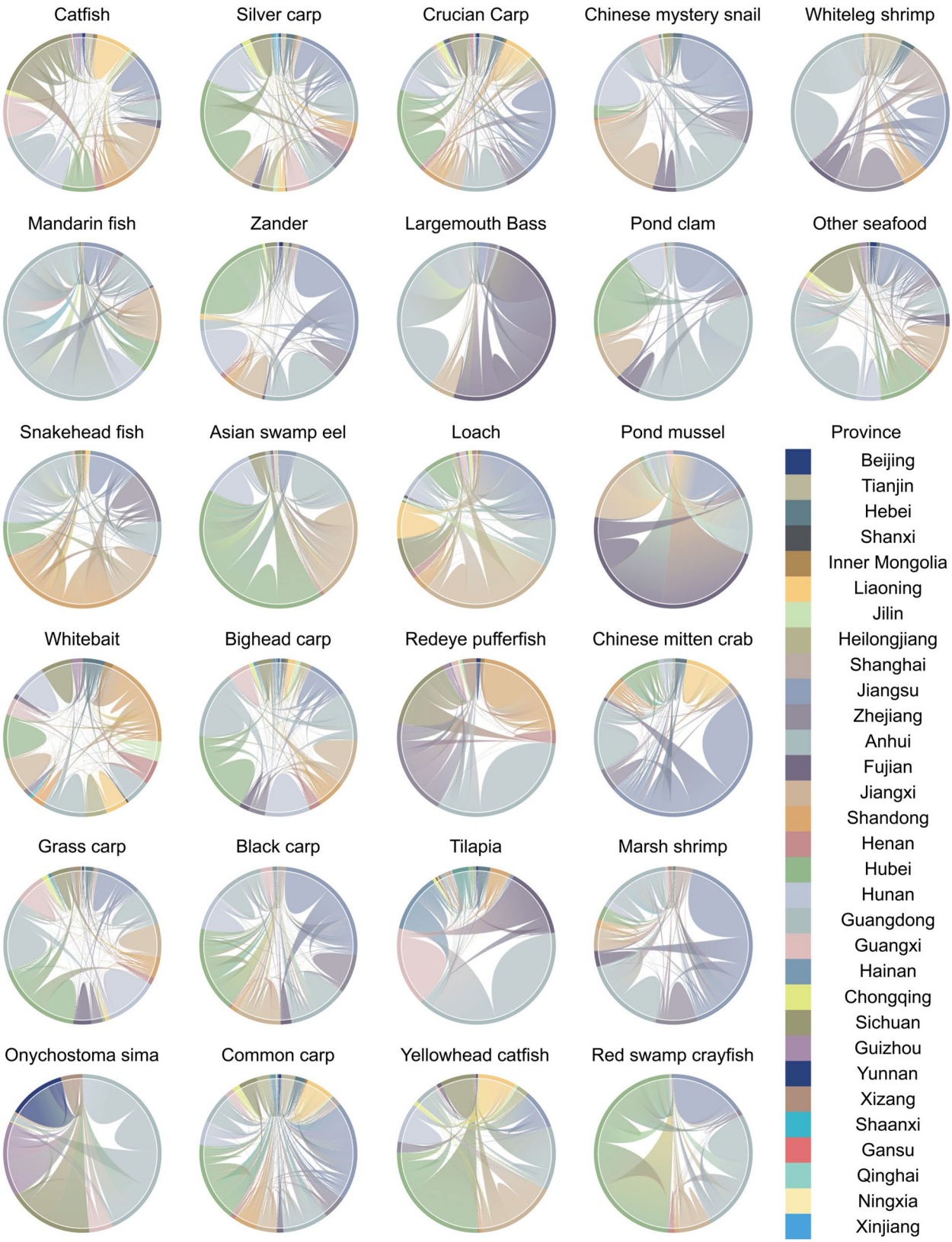
Extended Data Fig. 4 | Changes in MeHg PDI through freshwater seafood consumption and associated health benefits in China in 2022. a, Nationwide decreases of MeHg PDI (ng kg^{-1} d^{-1}), disaggregated by gender, type of residence, and age group. **b**, Provincial decreases of MeHg PDI (ng kg^{-1} d^{-1}). The yellow and green columns denote provincial IQ increase per fetus (point) and avoided deaths due to fatal heart attack in 2022, respectively. Base map in **b** from https://www.tianditu.gov.cn/.

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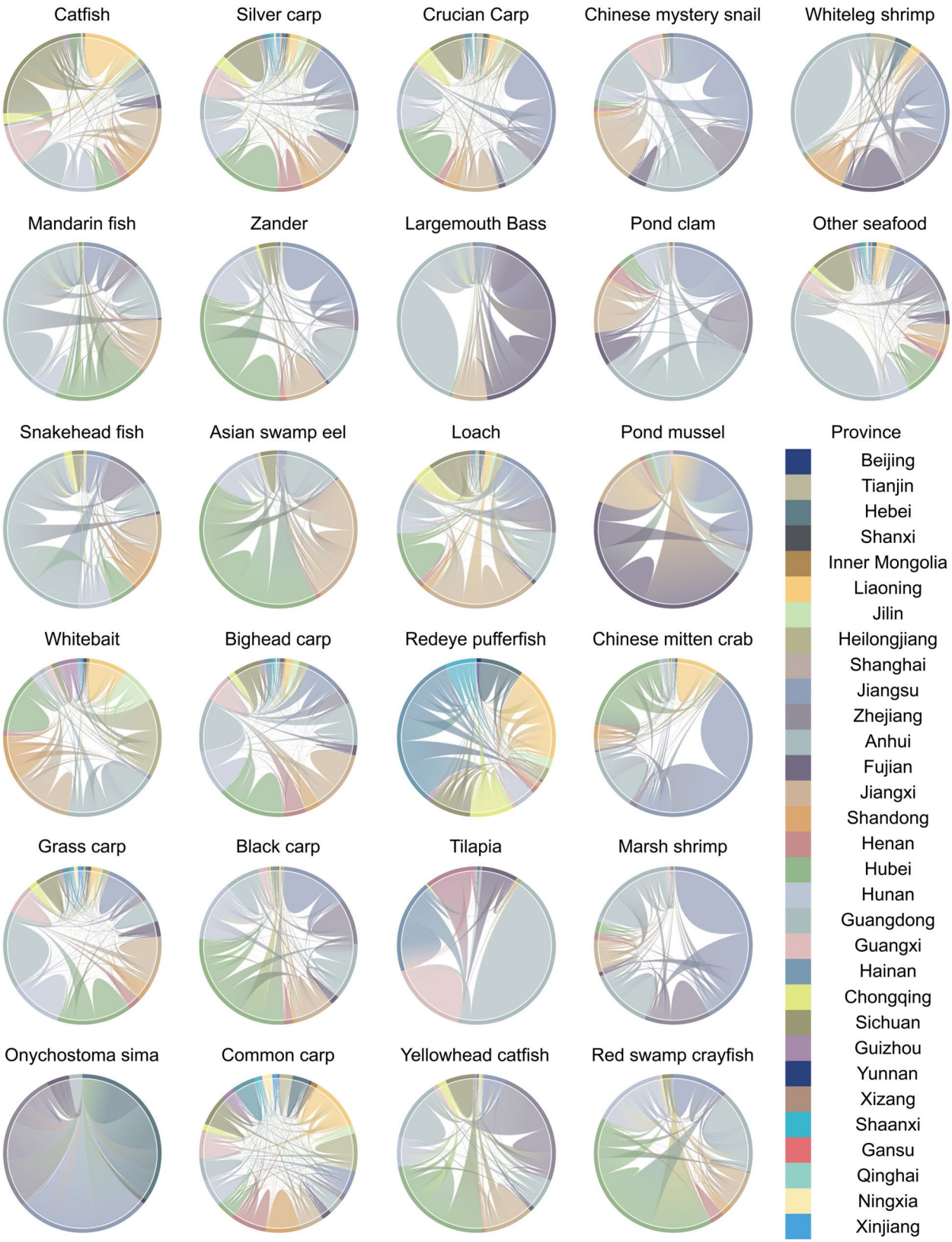
Extended Data Fig. 5 | Aquatic Hg concentration of sub-watersheds in China in the absence of MWT plants in 2022. According to China's Environmental quality standards for surface water[110], the Hg concentration limit for Class-I and Class-II waterbodies is 0.05 μg L^{-1}. The Hg concentration limit for Class-III waterbodies is 0.1 μg L^{-1}, which is the Hg concentration limit for aquatic breeding areas. The Hg concentration limit for Class-IV and Class-V waterbodies is 1 μg L^{-1}. Base map from https://www.tianditu.gov.cn/.

https://doi.org/10.1038/s43016-024-01093-9



Extended Data Fig. 6 | Interprovincial transport of common freshwater seafood species in 2002. Transportation routes with volumes less than 1 ton were not depicted in the figure.

https://doi.org/10.1038/s43016-024-01093-9



Extended Data Fig. 7 | Interprovincial transport of common freshwater seafood species in 2022. Transportation routes with volumes less than 1 ton were not depicted in the figure.

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Last updated by author(s): Oct 30, 2024

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Study description	This study established models to investigate the impact of the reduction of mercury releases of municipal wastewater treatment in China upon freshwater seafood safety and human health. Field experiment and sampling are not included in this study.
Research sample	Data on China's municipal wastewater treatment are sourced from National Urban Sewage Treatment Management Information System (wsxm.cin.gov.cn) and related official yearbooks (China Urban Construction Yearbook, China Environment Yearbook, and China Environmental Statistics Yearbook). Data on river reaches, watersheds, and precipitation in China are sourced from the HydroATLAS database (www.hydrosheds.org/hydroatlas) and Science Data Bank (www.scidb.cn/en/cstr/31253.11.sciencedb.01607). Data on China's seafood production, processing, consumption, and demographic data are sourced from relevant yearbooks in China (China Fishery Yearbook and China Fishery Statistics Yearbook, China Statistics Yearbook, and China Health Yearbook). Consumer price index data are referred from World Bank Open Data (data.worldbank.org/indicator/FP.CPI.TOTL). Intake data are referred from China Health and Nutrition Survey (www.cpc.unc.edu/projects/china).
Sampling strategy	This study does not rely on a sampling strategy.
Data collection	The authors collected the samples of mercury concentration of sewage sludge, mercury bioconcentration factor, mercury concentration in seafood and in water bodies from official datasets and relevent literature, as cited in Supplementary Information.
Timing and spatial scale	The study period is 1980-2022. The study domain is China.
Data exclusions	The study covered all data samples.
Reproducibility	All sources of input data, detailed calculation method, and model parameters are disclosed in main text and Supplementary Information.
Randomization	Randomization is not relevant to this study as no experiments or trials were undertaken.
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